

# Reducing Query Cost over a Frozen Document Index: What the Index Decides

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## Abstract

A dense document index is often the one component of a retrieval system that cannot be rebuilt: re-encoding is expensive, and every downstream consumer depends on the stored vectors. The query encoder has no such constraint. Query-side distillation trains a cheap student to emit queries in the frozen index’s space, so the index stays and only the query path changes. This paper asks what that substitution costs and what properties of the index decide the cost. We fit closed-form query students to 26 public embedding indexes under two student representations and ask, first, how much retrieval quality the student recovers and what predicts it. The index’s own retrieval quality predicts the student’s quality only once the index’s width is held fixed: standardized effects of +0.69 for quality and  $-0.83$  for width, and a two-variable model fitted on ten indexes ranks 16 held-out indexes at Spearman 0.85. The strongest index yields one of the weakest students. Second, we ask what the student costs at search time over the index’s unchanged HNSW graph: its queries recover fewer of their exact neighbors at equal effort in all 25 measured indexes on two corpora, and matching the index’s own queries takes a median four times the search budget. Constella, three compatible query encoders over one frozen Stella index, is the worked instance.

## 1. Introduction

A production retrieval system accumulates inertia in its document index. Encoding the 23.7 million documents of the BEIR-15 collection with a 400-million-parameter encoder took a measured 39.9 hours on one RTX 3080; a production index also carries its graph, its quantization, and every consumer that stores or joins against its vectors. For many teams the index is effectively immutable. The query encoder is the opposite: it runs once per request, it can be replaced between requests, and nothing stored depends on it, as long as its output lands in the index’s space.

**Query-side distillation** makes that condition hold. A student encoder is trained to reproduce the frozen encoder’s query vectors and then searches the frozen document vectors directly [QED, EmbedDistill, LEAF]. Static variants fit a token table instead of a transformer, so a query costs a table lookup and a sum [pyNIFE, LightRetriever]. The capability is established. What is not established is what it costs over a given index, and whether the cost can be predicted from the index before anything is built.

We study that question over 26 public embedding indexes. Because the students we use are fit in closed form in minutes, the same recipe can be applied to every index under one protocol, which turns a one-model demonstration into a comparison across indexes. Two questions organize the paper.

- **RQ1, retention.** How much retrieval quality can a closed-form query student recover from a frozen index, and which properties of the index predict it? The field’s working assumption, which we take as

the hypothesis to test, is that a better index yields a better student.

- **RQ2, search effort.** Over the index’s unchanged approximate-search graph, how much more search effort does the student’s query path need than the index’s own query path, and is that effort predictable?

The results are these. Against RQ1, the index’s own retrieval quality predicts the student’s quality only once the index’s width is held fixed. Width is the larger effect and works against the student; because stronger indexes are usually wider, the two effects cancel in a pooled comparison, and the strongest index in the roster yields one of the weakest students under both student representations. A two-variable model fitted on ten indexes ranks 16 others at Spearman 0.85 for one representation and 0.70 for the other, where the index’s own score alone ranks them at 0.37 and 0.16 [E24 prospective test pending]. Against RQ2, the student’s queries recover fewer of their own exact neighbors than the index’s queries at equal HNSW effort in all 25 measured indexes on two corpora of 57,638 and 171,332 documents, and matching the index’s own recovery takes a median four times the search budget; on a 25,657-document corpus the gap is smaller [E22 predictors and E23 size test pending].

**Constella** is the worked instance: a frozen Stella index [Stella] with three compatible query paths, Stella’s own encoder, a 34.5-million-parameter transformer student (Nano), and a token-table student (Zero), released with the registered evaluations that gated the release. Section 6 reports it.

Throughout, “registered” marks the project’s pre-registered evaluations, whose decision rules were fixed before any result was observed, and “exploratory” marks analyses added afterwards. The two index-wide screens that answer RQ1 and RQ2 are exploratory; their analyses were written down before scoring, and the prospective test of RQ1 commits its predictions before any student is fitted.

## 2. Related work

Query-side distillation against a frozen document encoder is established. QED retains 92.5% of a dense retriever’s BEIR score with a two-layer student trained by embedding alignment [QED]; EmbedDistill keeps the teacher’s document encoder and distills an asymmetric student [EmbedDistill]; LEAF aligns a small student to a larger teacher’s representation space and reports 97.7% retention at 4.7-fold compression [LEAF]. pyNIFE fits a static token table to a frozen teacher and proposes switching between the static and the contextual path [pyNIFE]; LightRetriever trains a lookup query side jointly with a document tower [LightRetriever]. Static sentence embeddings and Model2Vec build table representations with their own indexes [StaticEmb, Model2Vec]; NanoVDR distills a vision-language retriever into a small text encoder [NanoVDR]. None of these compares students across many teachers under one recipe, which is the comparison RQ1 needs.

That stronger teachers can produce weaker students is known in distillation generally [Cho and Hariharan 2019] and in dense retrieval [PROD]; the explanations offered there concern capacity gaps between teacher and student. Our contribution to that line is a measured competing explanation, the teacher’s width, with a held-out prediction.

Graph-based approximate search over HNSW is sensitive to where queries sit relative to the indexed vectors [HNSW]; out-of-distribution queries are known to need more effort, and graph constructions exist that use query samples to mitigate it [OOD-DiskANN, RoarGraph]. RQ2 measures that effect for queries produced by a student of the same index, which are near the index’s distribution but not on it. Query-aware decoding of binary sketches is the precedent for the precision control in Section 5 [QA-Cos].

## 3. Experimental setup

### 3.1 Indexes

The roster is 26 public English embedding models that share the 30,522-entry BERT WordPiece vocabulary, a requirement of the table student. Ten were registered before any result; 16 were added as an exploratory expansion under the same recipe and selection order. Each model is an index in the sense of RQ1: its document vectors are frozen and its own query encoder is the reference path. Table A lists them by width.

**Table A.** The 26 indexes. “Own nDCG@10” is the index’s own query path on the six-set suite (Section 3.3); the two student columns are the outcomes of Section 4.

| Index<br>(teacher)   | Width | Pooling | Own<br>nDCG@10 | Table student | Head student | Roster      |
|----------------------|-------|---------|----------------|---------------|--------------|-------------|
| bge-small-en-v1.5    | 384   | cls     | 0.5042         | 0.3581        | 0.5084       | exploratory |
| arctic-embed-s       | 384   | cls     | 0.4993         | 0.3516        | 0.2594       | exploratory |
| gte-small            | 384   | mean    | 0.4838         | 0.2903        | 0.4733       | exploratory |
| arctic-embed-xs      | 384   | cls     | 0.4662         | 0.3440        | 0.2779       | exploratory |
| e5-small-v2          | 384   | mean    | 0.4544         | 0.3203        | 0.3367       | exploratory |
| minilm-l12           | 384   | mean    | 0.4219         | 0.3138        | 0.3229       | exploratory |
| minilm-l6            | 384   | mean    | 0.4142         | 0.3267        | 0.3088       | exploratory |
| multi-qa-minilm-l6   | 384   | mean    | 0.4008         | 0.3406        | 0.3094       | exploratory |
| msmarco-minilm-l6    | 384   | mean    | 0.3281         | 0.2905        | 0.1962       | exploratory |
| gte-base-en-v1.5     | 768   | cls     | 0.5331         | 0.3252        | 0.2618       | registered  |
| arctic-embed-m-v1.5  | 768   | cls     | 0.5263         | 0.3279        | 0.1874       | registered  |
| bge-base-en-v1.5     | 768   | cls     | 0.5259         | 0.3529        | 0.3249       | registered  |
| bge-base-en-v1       | 768   | cls     | 0.5131         | 0.3004        | 0.2773       | exploratory |
| gte-base             | 768   | mean    | 0.5063         | 0.2762        | 0.2314       | exploratory |
| e5-base-v1           | 768   | mean    | 0.4934         | 0.3126        | 0.2345       | exploratory |
| e5-base-v2           | 768   | mean    | 0.4669         | 0.2930        | 0.2241       | registered  |
| contriever-msmarco   | 768   | mean    | 0.3909         | 0.3270        | 0.1879       | exploratory |
| tas-b                | 768   | cls     | 0.3262         | 0.2115        | 0.0868       | exploratory |
| contriever           | 768   | mean    | 0.2846         | 0.1864        | 0.0590       | exploratory |
| gte-large-en-v1.5    | 1024  | cls     | 0.5970         | 0.2455        | 0.2229       | registered  |
| stella-400M-v5       | 1024  | mean    | 0.5745         | 0.3974        | 0.2505       | registered  |
| mxbai-embed-large-v1 | 1024  | cls     | 0.5368         | 0.2605        | 0.2641       | registered  |
| bge-large-en-v1.5    | 1024  | cls     | 0.5329         | 0.2845        | 0.2679       | registered  |
| arctic-embed-l       | 1024  | cls     | 0.5289         | 0.3034        | 0.1564       | registered  |
| gte-large            | 1024  | mean    | 0.5129         | 0.2481        | 0.2249       | exploratory |
| e5-large-v2          | 1024  | mean    | 0.4735         | 0.2585        | 0.1743       | registered  |

The roster is related: nine model families, three width levels, and several checkpoints that are revisions of one another. Checkpoint-level intervals in this paper resample checkpoints and do not treat them as independent; family-level sensitivity is reported beside them. One control, arctic-embed-l read out with mean pooling instead of its published CLS pooling, is excluded from every roster statistic.

### 3.2 Student representations

Two closed-form students are fitted to every index, each producing a 1024-or-narrower query vector in the index’s space.

**Table student.** A matrix of one row per WordPiece token, fitted by ridge regression from the token counts of each fit query to the index’s query vector for that query, initialized from the index’s own token embeddings and solved by block conjugate gradient with a convergence gate. A query is the normalized, square-root-count-weighted sum of its tokens’ rows; no transformer runs at query time.

**Head student.** A frozen bge-small backbone, read out as the concatenated masked-mean states of layers 12, 8, and 4 (1,152 dimensions) plus a bias, with a dense ridge head fitted to the index’s query vectors; the output is normalized. The backbone is the feature path of the Nano instance. It is a second representation under the same screening setup, chosen to test whether RQ1’s answer depends on the table.

Both students are floors for a trained student of the same architecture; neither selects the trained Zero or Nano of Section 6.

### 3.3 Fit data, selection, and scoring

Both students are fitted on the same 337,981 screened fit queries, encoded by each index’s own query path with its published prompt. The ridge penalty is chosen per index on two development forums (CQADupStack physics and programmers) and frozen before any public dataset is scored. Retrieval is scored by exact search, nDCG@10 with the registered self-hit drop, on six public BEIR datasets: NFCorpus, SCIDOCs, SciFact, TREC-COVID, FiQA, and ArguAna. The six-set macro is the outcome; the four sets without disclosed exposure for the Stella instance (the first four) are a sensitivity outcome. Each index is scored once per set.

### 3.4 Approximate-search protocol

RQ2 uses Qdrant 1.19.1 with HNSW  $m=16$ ,  $ef\_construct=100$ , every segment indexed, cosine distance, no quantization. For each index and each of three workloads, FiQA (57,638 documents), SCIDOCs (25,657), and TREC-COVID (171,332), two graphs are built with different seeded insertion orders, and  $ef$  is swept over 16, 32, 64, 128, 256, and 512 for the index’s own queries, the table student’s queries, and the head student’s queries. Each path is measured against its own exact top-10 over the same document vectors. Recovery@10 is tie-aware: a returned document counts when its exact score is at or above the exact tenth neighbor’s score within  $1e-4$ , because some table students produce exact-score ties at rank ten. Exact parity between the engine and a NumPy reference is checked on 200 queries per path and build; one index, tas-b, could not reach parity because its rank-10 score gaps lie at float32 precision, and it is excluded from RQ2. The pre-specified primary quantity is the relative-nDCG-loss multiplier: with the index’s own path at  $ef=64$  as the reference, the smallest  $ef$  at which the student path reaches the same relative loss, divided by 64, averaged over the two builds, and censored when a build never reaches it by  $ef=512$ . The secondary quantity is the same multiplier on recovery@10, and the recovery gap at  $ef=64$ .

### 3.5 The instance

Constella is built around Stella, the `stella_en_400M_v5` encoder, pinned revision `ffeb2b7e`, whose 1024-dimensional normalized document vectors are frozen throughout. Zero is a trained table of 30,522 rows exported as int8; Nano is a 34.5-million-parameter transformer on the head student’s feature path, trained with a squared-error loss against Stella’s query vectors. Both are evaluated on BEIR-15 by exact search and timed under a registered protocol on an Apple M5 Pro CPU. Their registered contrasts and the one-shot held-out evaluation are in Appendix A.

## 4. RQ1: what the index decides about retention

### 4.1 Hypothesis and the pooled result

**H1.** The stronger the index’s own retrieval, the stronger the closed-form student fitted to it; in particular, the strongest index yields the strongest student.

H1 is the heuristic a team would use to choose which index to build a cheap query path for, and it is what the project used before the screen. Table A already contains its refutation in the pooled form. The strongest registered index, gte-large-en-v1.5 at 0.5970, yields a table student of 0.2455, the weakest of the registered ten, and a head student of 0.2229; Stella at 0.5745 yields the strongest table student at 0.3974. Over all 26 indexes, the rank correlation between an index’s own score and its student’s score is +0.09 for the table (95% checkpoint interval  $-0.39$  to  $+0.52$ ) and +0.09 for the head ( $-0.36$  to  $+0.51$ ). The intervals admit a moderate positive association; they exclude the strong one H1 asserts.

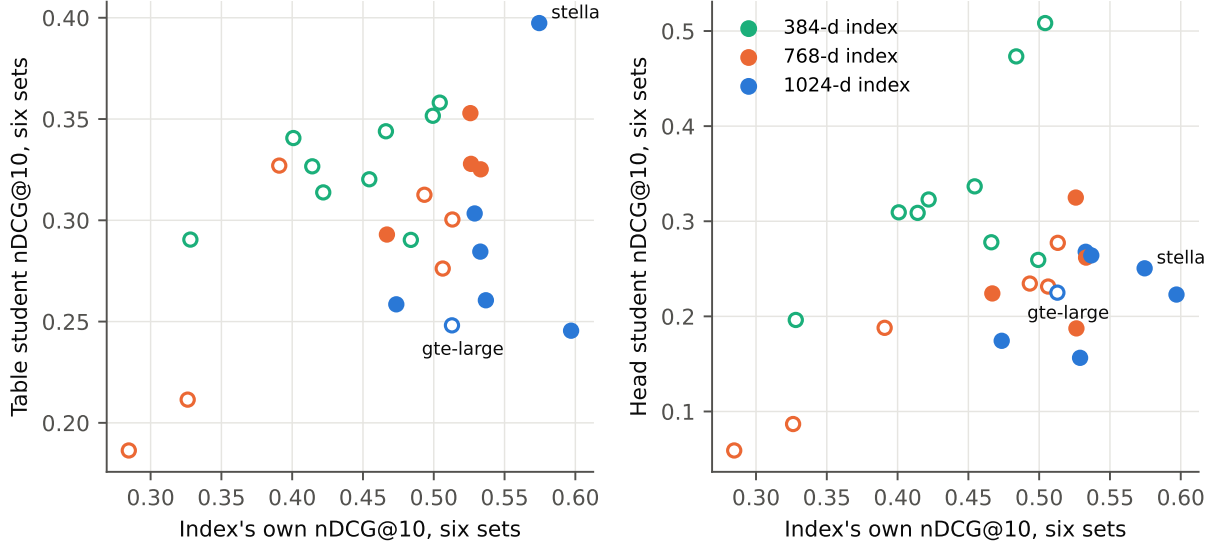


Figure 1: Student quality against the index’s own quality for the table (left) and head (right) students, colored by index width. Filled points are registered indexes.

## 4.2 Width as the competing explanation

Figure 1 shows the pooled null is two opposing effects. Within each width level, student quality rises with index quality; across levels, wider indexes yield weaker students. Table B fits that structure. The outcome is the student’s absolute six-set score; the predictors are the index’s own score and  $\log_2$  width; coefficients are standardized, with 10,000-draw checkpoint bootstrap intervals; held-out performance is leave-one-family-out prediction over the nine families.

**Table B.** Nested models of student quality over 26 indexes (exploratory, analysis pre-specified).

| Student | Predictors              | Index quality<br>beta   | Width beta              | In-sample $R^2$ | Leave-one-family-<br>out Spearman |
|---------|-------------------------|-------------------------|-------------------------|-----------------|-----------------------------------|
| Head    | index quality           | +0.34 [−0.16,<br>+0.65] |                         | 0.12            | −0.17                             |
| Head    | index quality,<br>width | +0.69 [+0.40,<br>+0.97] | −0.83 [−1.14,<br>−0.60] | 0.70            | +0.69                             |
| Table   | index quality           | +0.37 [−0.30,<br>+0.72] |                         | 0.14            | −0.15                             |
| Table   | index quality,<br>width | +0.63 [+0.06,<br>+0.89] | −0.64 [−0.95,<br>−0.32] | 0.48            | +0.40                             |

Index quality and width correlate at +0.61 across the roster, which is why the pooled correlation is near zero. The partial rank correlation between student and index quality given width is +0.58 for the head and

+0.45 for the table. Adding nine family indicators raises in-sample  $R^2$  to 0.91 and 0.81 and leaves the signs unchanged; those models cannot be validated out of family and are reported only in sample. The width effect is the larger of the two under both representations.

### 4.3 Held-out and prospective prediction

The roster’s own history supplies a held-out test. The two-variable model fitted on the ten registered indexes, scored first, ranks the 16 indexes scored later at Spearman 0.85 [0.53, 0.99] for the head and 0.70 [0.20, 0.96] for the table; the index’s own score alone ranks them at 0.37 and 0.16. The hypothesis in Table B was formed after all 26 were seen, so this is a held-out fit rather than a prospective test. [E24 pending: eight further indexes never scored before, with predictions committed by hash before any student was fitted.]

### 4.4 Selection regret and the student’s own screen

**Table C.** Six-set nDCG@10 lost against the best student in the roster, by selection rule.

| Rule                                      | Head: pick, regret       | Table: pick, regret      |
|-------------------------------------------|--------------------------|--------------------------|
| Strongest index (H1)                      | gte-large-en-v1.5, 0.285 | gte-large-en-v1.5, 0.152 |
| Two-variable model, leave-one-out<br>pick | arctic-embed-s, 0.249    | gte-small, 0.107         |
| Student’s own dev-forum screen            | gte-small, 0.035         | stella-400M-v5, 0.000    |

The model explains the structure but does not make the pick: its leave-one-out choice still loses 0.25 and 0.11. The student’s own development screen, two forums scored with the fitted student, predicts the six-set rank at Spearman 0.84 (head) and 0.88 (table) and makes the best or near-best pick. Added to the two-variable model it dominates (standardized +0.71 [+0.36, +0.99] for the head) and raises leave-one-family-out prediction to 0.77 and 0.80. Width is what the index decides; the screen is how a team should decide.

One caveat governs Section 4: 26 related indexes at three width levels support a hypothesis about what a closed-form student pays for width, not a mechanism, and the head’s own backbone appears in the roster (bge-small-en-v1.5, head retention 1.008); every statistic above changes by less than 0.06 without it.

## 5. RQ2: what the index decides about search effort

### 5.1 Hypothesis

**H2.** Over the index’s unchanged HNSW graph, the student’s query path needs more search effort than the index’s own query path to recover the same fraction of its exact neighbors, because the student’s queries sit differently against the indexed vectors.

The first half of H2 is testable per index; the second half is the mechanism, tested in Section 5.3.

### 5.2 Effort across 25 indexes

**Table D.** Search effort of the table student relative to the index’s own path at **ef=64**, two builds averaged, 25 indexes (exploratory, analysis pre-specified). The primary quantity is the relative-nDCG-loss multiplier; “censored” means at least one build never reached the reference by **ef=512**.

| Workload   | Documents | Loss multiplier:<br>indexes<br>above 1 /<br>at or<br>below | Censored | Loss multiplier<br>median over<br>reached | Recovery gap at<br>ef=64,<br>mean<br>(indexes<br>below 0) | Recovery multiplier<br>median |
|------------|-----------|------------------------------------------------------------|----------|-------------------------------------------|-----------------------------------------------------------|-------------------------------|
| FiQA       | 57,638    | 25 / 0                                                     | 4        | 4.0                                       | -2.9 pp<br>(25 of 25)                                     | 4.0                           |
| SCIDOCS    | 25,657    | 17 / 8                                                     | 8        | 1.25                                      | -0.8 pp<br>(17 of 25)                                     | 2.0                           |
| TREC-COVID | 171,332   | 12 / 13                                                    | 8        | 0.25                                      | -3.9 pp<br>(25 of 25)                                     | 4.0                           |

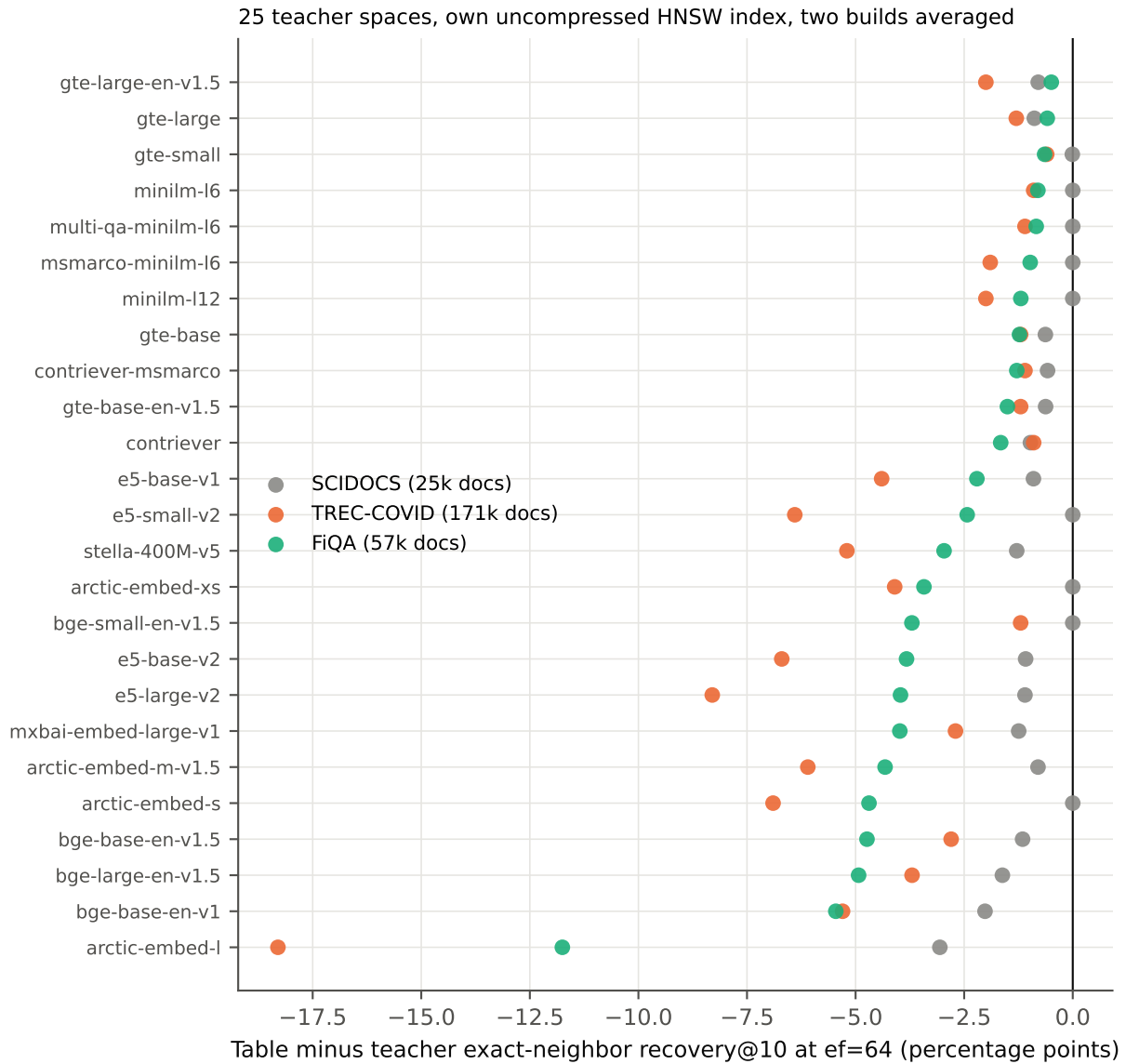


Figure 2: Recovery gap at ef=64 by index, three workloads, two builds averaged.

On FiQA every index needs more effort by the primary measure, with a median of four times the reference **ef** over the 21 reached and four censored above eight. On SCIDOCS 17 indexes need more and eight do not. On TREC-COVID the primary measure is mixed, 12 above and 13 at or below, with a median of 0.25; its index paths lose only 0.2% of nDCG at **ef**=64 over 50 queries, so a threshold on relative loss there is sensitive to differences smaller than its noise. The secondary measure is consistent across all three: the student recovers fewer of its own exact neighbors at **ef**=64 in 25 of 25 indexes on FiQA and TREC-COVID and in 17 of 25 on SCIDOCS, with a median recovery multiplier of four on the two larger corpora and two on the smallest. The head student shows the same recovery deficit (24 and 23 of 25). The first half of H2 holds for recovery on every corpus and for relevance on the largest-query corpus.

### 5.3 Is the effort predictable?

Two summaries of query placement measured in the Stella instance, the median cosine of a query to its nearest document and the median gap between its first and tenth scores, differ between student and index paths in every roster (the student’s queries sit farther from the documents, with flatter margins). Across the 25 indexes they do not predict the size of the gap: Spearman 0.00 on FiQA, +0.23 on SCIDOCS, and −0.34 on TREC-COVID, with intervals spanning zero. [E22 pending: Table E, declared placement features (query-to-document distance ratio, hubness, top-10 overlap, effective rank) with clustered intervals and leave-family-out prediction.]

### 5.4 Corpus size

The gap is smallest on the smallest corpus, where the index’s own path already loses almost nothing at **ef**=64. Three workloads confound corpus size with domain. [E23 pending: FiQA subsampled to 25,657 documents on the same queries, paired across indexes.]

One caveat governs Section 5: one engine, one graph configuration, closed-form students, and timing on a shared machine; a multiplier on **ef** is a multiplier on graph effort, not on latency, and a recovery deficit is not a relevance deficit until the primary measure agrees.

## 6. The Constella instance

Over Stella’s frozen index, the three query paths give the budgets in Table 1, by exact search over BEIR-15.

**Table 1.** Three query paths over one frozen Stella index. Encoding medians: 100 real 5-to-12-word queries, batch one, four CPU threads, Apple M5 Pro, registered protocol.

| Query path               | Exact nDCG@10 | Share of Stella | Encode p50 | Assets     |
|--------------------------|---------------|-----------------|------------|------------|
| Stella query encoder     | 0.5614        | reference       | 31.6 ms    | 1669.6 MiB |
| Nano (34.5M transformer) | 0.5081        | 90.5%           | 2.25 ms    | 132.3 MiB  |
| Zero (token table)       | 0.4572        | 81.4%           | 0.044 ms   | 90.1 MiB   |

Retention varies by dataset, from 0.67 for Zero on TREC-COVID and FiQA to 0.96 on climate-fever and Quora (Figure 3). RQ1’s answer applies to this instance: Stella is a 1024-wide index with the best own score among its width band, and its table student is the roster’s best, while a wider-and-stronger index would not have been.

RQ2’s answer applies too. On the uncompressed million-passage diagnostic, search time at a fixed **ef** is the same for all three paths within 5%, and the paths differ in the **ef** they need: to stay within 1% of their own exact score, Stella needs **ef**=128, Nano 256, and Zero 512, with search medians of 1.53, 2.18, and 3.58 ms. On one binary-quantized collection the fastest setting within 1% gives Zero 0.6116 at 1.11 ms, Nano 0.6893 at 2.48 ms, and Stella 0.7226 at 21.3 ms: a 63-fold encoding gap between Zero and Nano becomes 2.2-fold after search, at different quality levels, and the part specific to the cheap path is the extra **ef**. Figure 4 gives the registered sweep. Fusing Zero with BM25 recovers more quality than any table-side tuning did (BEIR-15 macro 0.4572 to 0.4933) at the cost of a second index and retrieval; per-query routing between

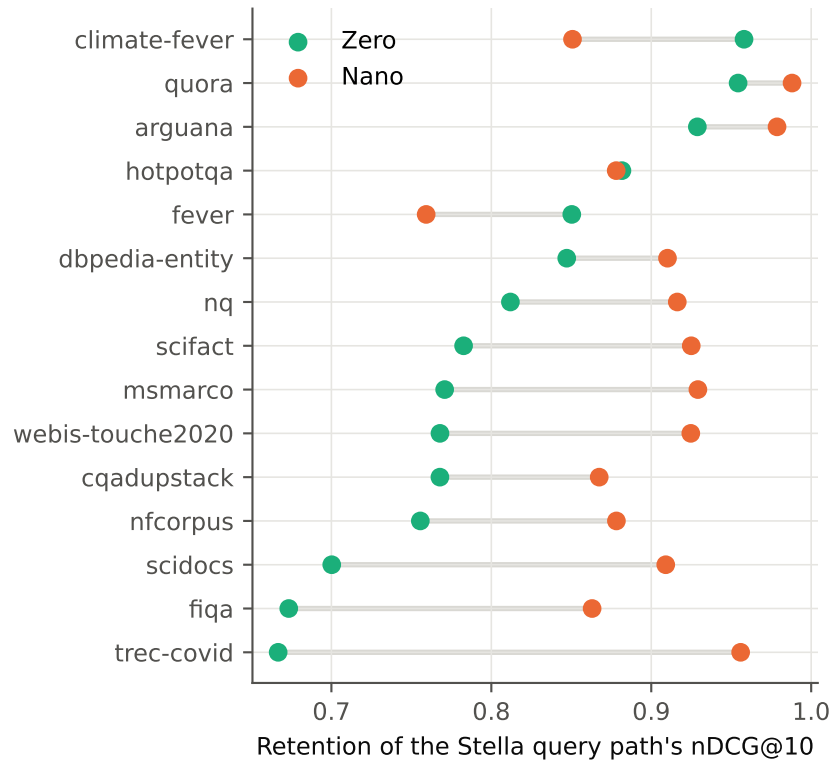


Figure 3: Retention per dataset. Datasets with disclosed Stella training exposure are ArguAna, FiQA, and FEVER; the four originally held-out sets are FEVER, DBpedia-entity, and the CQADupStack android and english forums.

Zero and Nano has an oracle headroom of 0.031 that the cheap selectors tested recover little of (Appendix D). With document codes fixed and no graph, scoring with full query magnitudes instead of signs raises Zero’s exact-top-10 coverage from 82.2% to 93.0% on FiQA at a 40-candidate budget, in line with query-aware decoding of binary sketches [QA-Cos]; the engine’s native setting for it was not tested (Appendix C).

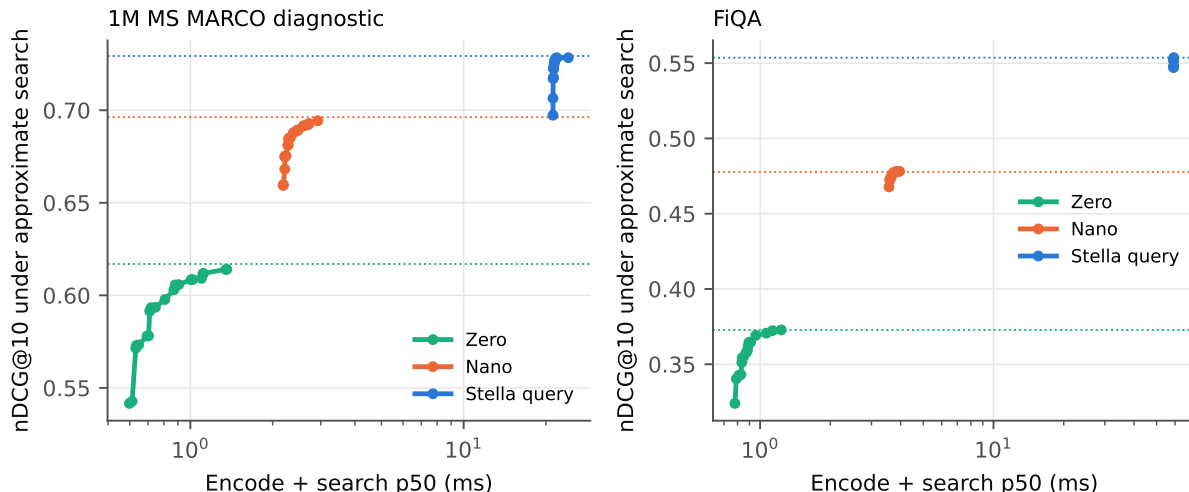


Figure 4: Quality under approximate search against encoding plus search time, registered sweep. Dotted lines are each path’s exact score.

## 7. Limitations

RQ1 rests on two closed-form students over 26 related indexes at three width levels; the students are floors for trained students of the same architecture, the screen does not select the trained Zero or Nano, checkpoint intervals do not cover training variance, and width covaries with family and quality. RQ2 rests on one engine and one graph configuration, three workloads that confound corpus size with domain, and 50 queries on TREC-COVID; its timing ran on a shared machine and is not a latency claim. The instance’s two students have different training pools, including different FEVER exposure, and Stella discloses training exposure on ArguAna, FiQA, and FEVER. Query intervals condition on the trained models. The native magnitude-preserving query setting was not tested.

## Artifact availability

The companion repository holds inference examples, build configurations, methods with their dated amendments, immutable result receipts, per-dataset tables, and the full research history including negative results. The released artifacts are Constella Zero, Constella Nano, and the Stella document encoder on the Hugging Face Hub. All released paths run in fp32 through the published loader; a candidate fp16 export passed CPU parity and failed on CUDA (minimum cosine 0.662), which is why qualification runs through the actual loader and device. Two inputs are identified by hash but not yet archived in the repository: the cleaned 337,981-query fit list and the sampled one-million-passage identifier list.

## Appendix A. Registered evaluations of the instance

**Serving configuration.** Stella is pinned to revision ffeb2b7e, with its s2p\_query prompt, fp32 encoding, normalized output, fp16 stored document vectors, and cosine similarity; maximum sequence length 512. Nano uses bge-small layers 12, 8, and 4 concatenated to 1152 dimensions before projection; batches mix 75% queries and 25% documents. Zero’s training pool includes FEVER-train and HotpotQA; Nano excludes FEVER. MS MARCO is licensed for non-commercial use, so it was validation-only in our distillation: it supplied no training targets, negatives, or generation seeds, although Nano’s pretrained base has prior exposure to it.

**Partitions.** The six-set suite is NFCorpus, SCIDOCs, SciFact, TREC-COVID, FiQA, and ArguAna. “Clean-four” means the first four, which have no disclosed Stella overlap; it is not a causal contamination estimate. Four datasets were held out and evaluated once: FEVER, DBpedia-entity, and the CQADupStack android and english forums. Two other forums, physics and programmers, select table penalties and routing settings and were reused during model construction.

**Reference systems.** bge-small-en-v1.5 and LEAF were the project’s selected targets, not a state-of-the-art frontier. They use their own document representations. On BEIR-15 they score 0.5171 and 0.5402 against Nano’s 0.5081, at 2.57 and 1.39 ms median encoding; BM25 scores 0.4006. Nano is below both on the broad average even though it passed some of the registered contrasts below.

**Nano’s registered contrasts** used a fixed sequence with one-sided 2.5% lower bounds. “Established” means the bound cleared zero; “unresolved” means it did not; “descriptive” rows were registered as reports, not tests. Intervals resample queries with the models fixed.

| Contrast         | Partition                           | nDCG@10 difference           | Outcome     |
|------------------|-------------------------------------|------------------------------|-------------|
| Nano – bge-small | clean-four                          | +0.0176; lower bound +0.0037 | established |
| Nano – bge-small | all six                             | +0.0274; lower bound +0.0173 | established |
| Nano – LEAF      | all six                             | +0.0162; lower bound +0.0065 | established |
| Nano – LEAF      | clean-four                          | −0.0011; lower bound −0.0145 | unresolved  |
| Nano – bge-small | held-out NDO-3,<br>dataset weighted | +0.0032 [−0.0069, +0.0134]   | descriptive |
| Nano – bge-small | held-out NDO-3,<br>query pooled     | −0.0121 [−0.0219, −0.0024]   | descriptive |
| Nano – LEAF      | held-out NDO-3                      | −0.0389 [−0.0488, −0.0290]   | descriptive |

NDO-3 is the registered held-out aggregate: DBpedia-entity weighted one half, the android and english forums one quarter each, FEVER excluded as the double-exposure sensitivity row.

**Zero’s registered contrasts** used a Holm correction across the family. Zero missed its dense bar and did not confirmatorily beat BM25.

| Contrast, all six                    | nDCG@10 difference | Raw 95% interval   | Outcome                                 |
|--------------------------------------|--------------------|--------------------|-----------------------------------------|
| Zero – LightRetriever<br>dense       | −0.0243            | [−0.0405, −0.0086] | below the 0.4583 bar                    |
| Zero – BM25                          | +0.0165            | [+0.0017, +0.0311] | unresolved; p=.0149<br>fails Holm .0083 |
| Zero + BM25 (convex)<br>– OpenSearch | +0.0043            | [−0.0063, +0.0151] | unresolved                              |

The registered clean-four sensitivity is descriptive: −0.0443 [−0.0675, −0.0212], −0.0311 [−0.0517, −0.0109], and −0.0107 [−0.0262, +0.0043]. Unresolved superiority does not establish equivalence. The confirmatory fusion operator was convex score fusion with development-fitted weight 0.8 at depth 1000; the later DBSF-at-100 recommendation is a deployability choice, not a new confirmation. On the development components DBSF minus convex is +0.0035 at 10 candidates, −0.0020 at 50, −0.0063 at 100, and −0.0146 at 1000.

**The held-out four**, copied from their one-shot published aggregate:

| System | FEVER  | DBpedia | CQADup android | CQADup english |
|--------|--------|---------|----------------|----------------|
| Nano   | 0.6231 | 0.4190  | 0.4774         | 0.4298         |
| Zero   | 0.6978 | 0.3900  | 0.4431         | 0.3690         |
| Stella | 0.8207 | 0.4603  | 0.5275         | 0.5183         |

| System      | FEVER  | DBpedia | CQADup android | CQADup english |
|-------------|--------|---------|----------------|----------------|
| bge-small   | 0.8662 | 0.4003  | 0.4761         | 0.4558         |
| LEAF        | 0.8671 | 0.4520  | 0.5260         | 0.4707         |
| BM25        | 0.5036 | 0.3045  | 0.3952         | 0.3481         |
| Zero + BM25 | 0.7559 | 0.4030  | 0.4629         | 0.4016         |
| Nano + BM25 | 0.6879 | 0.4296  | 0.4750         | 0.4323         |

## Appendix B. Construction details

**Table student.** Ridge regression from WordPiece token counts to the index’s query vectors, initialized from the index’s own token embeddings, solved by block conjugate gradient with a convergence gate; penalty grid 1e-4 to 1e-1 with one decade of extension at an edge, chosen on the two development forums and frozen before six-set scoring. The fit list was cleaned after an earlier version had 1.31% held-out-query overlap. One checkpoint failed the convergence gate and is excluded. Registered checkpoint exposure is documented; the exploratory expansion was not individually audited.

**Head student.** Frozen bge-small at the Nano build’s pinned revision; masked-mean-pooled hidden states of layers 12, 8, and 4 concatenated with a bias column; dense ridge with penalty scaled by the feature Gram trace, same grid, same selection order, same freeze; output normalized. bge-small-en-v1.5 is also an index in the roster and shares the backbone, and gte-small is nearly its linear image (retention 1.008 and 0.978); correlations without the backbone are +0.09 against the index and +0.46 against the table.

**Correlation rosters.** Registered ten: Spearman head-versus-index +0.18 (−0.56 to +0.67), head-versus-table +0.24 (−0.49 to +0.74), head development-versus-public +0.71 (+0.12 to +0.96). Leave-one-family-out values range from +0.44 to +0.60 for head-versus-table and −0.12 to +0.27 for head-versus-index over the 26. Within width bands, head-versus-index Spearman is +0.47 [−0.47, +1.00] at 384 (n=9), +0.67 [−0.03, +1.00] at 768 (n=10), and +0.36 [−0.76, +0.87] at 1024 (n=7).

**Fidelity is not retrieval.** Tables fitted to two e5 checkpoints match their index’s query vectors at mean cosine 0.90 and 0.89 against 0.78 for Stella’s table, yet retrieve worse; three earlier training screens reduced their losses without improving ranking. Three vector-error diagnostics did not rank table quality either.

**Costs.** Table fitting four to seven minutes per registered index on an A100, including query encoding except cached Stella targets; head features 43 seconds for the fit list, dev grid 14 seconds per index, six-set scoring about one minute per index. Nano’s final optimization ran 57.3 hours on one A100 for 199,999,721 example presentations, about \$95 at the recorded rental rate, excluding target preparation, data generation, recipe search, failed runs, and evaluation. Zero’s retraining was estimated at 20 minutes with prepared targets, plus 8 to 12 hours to encode targets for a new index.

## Appendix C. Search and precision controls

**Instance sweep.** Qdrant 1.19.1, HNSW m=16, ef\_construct=100, indexing threshold 1 KB so no segment is served by a plain scan, ef in 16 to 512, oversampling 1, 2, and 4 under compression, exact parity against NumPy on 200 queries per path. Timing is warmed and sequential with encoding and search measured in separate phases. At ef=16 the uncompressed million-passage losses are 4.4%, 7.0%, and 16.3% for Stella, Nano, and Zero; under binary quantization without oversampling 6.1%, 9.2%, and 19.8%.

**Cross-index sweep.** The same engine and graph parameters, uncompressed, two builds per index and workload differing in a seeded insertion order, 16 to 512 ef, limit 11 with the self-hit drop, tie-aware exact parity on 200 queries per path. Four amendments were recorded during the run, before any index’s rows were analysed: SCIDOCS added as a third workload because TREC-COVID has 50 queries; the tie-aware measure; a 0.995 sanity gate on parity with all values reported; and the exclusion rule that removed tas-b (SCIDOCS head parity 0.994). Stella’s own-path rows agree with the instance sweep at every ef. Loss-based censoring: four indexes on FiQA, eight on SCIDOCS, eight on TREC-COVID; recovery-based censoring: one per workload.

**Native binary pilot and replication.** On one fixed binary collection per workload with 192 deterministically selected queries, binary scoring with rescoring at 1x oversampling loses 11.7, 5.1, and 2.3 percentage points of exact-neighbor recovery for Zero, Nano, and Stella on FiQA at `ef=64`; the million-passage replication loses 3.9, 3.1, and 2.2, and the extra Zero-versus-Nano interval includes zero. Native rescoring also changes cross-segment candidate merging.

**Graph-free precision control.** Same queries, vectors, and targets; all document sign codes scored exhaustively at global candidate budgets 10, 40, and 100; expected coverage averages uniform inclusion at tied boundaries.

| Workload      | Path   | Sign query | Full query | Gain, points |
|---------------|--------|------------|------------|--------------|
| FiQA          | Zero   | 0.8217     | 0.9297     | +10.80       |
| FiQA          | Nano   | 0.9137     | 0.9771     | +6.33        |
| FiQA          | Stella | 0.9638     | 0.9922     | +2.84        |
| 1M diagnostic | Zero   | 0.9010     | 0.9677     | +6.67        |
| 1M diagnostic | Nano   | 0.9484     | 0.9833     | +3.49        |
| 1M diagnostic | Stella | 0.9685     | 0.9932     | +2.47        |

The paired extra Zero gains over Nano are 4.46 points on FiQA (95% query interval 2.53 to 6.43) and 3.17 on the diagnostic (1.58 to 4.75). Zero starts with more misses and does not repair a larger fraction of them. Full query values are not native scalar8 encoding, and global candidate counts are not per-segment oversampling.

## Appendix D. Fusion and routing over the frozen index

**Lexical fusion.** Fusing Zero with BM25 raises its BEIR-15 macro score from 0.4572 to 0.4933, 87.9% of Stella; Nano plus BM25 reaches 0.5110. The sign varies by workload: Zero on HotpotQA rises from 0.6127 to 0.7015, Nano on MS MARCO falls from 0.4063 to 0.3699. On the million-passage diagnostic a dense-plus-sparse collection takes 4.71 ms per query including Zero encoding against 1.11 ms for the dense binary example, in different configurations, not an isolated fusion overhead. The scores use distribution-based score fusion over 100 candidates per branch with a bm25s lexical branch [DBSF]; other lexical implementations need their own comparison because their score distributions enter the normalization. The operator ordering changes with candidate depth (Appendix A).

**Routing and blending (registered).** On 12 datasets, an oracle choosing Zero or Nano per query with relevance labels scores 0.5473 against 0.5161 for always-Nano, and matches always-Nano by sending the most beneficial 15% of queries to Nano. Query length, subwords per word, and the pooled vector norm are weak selectors (0.002 to 0.005 over random routing at matched Nano share); Zero’s post-search score margin recovers about a quarter of the oracle gain over random routing on six datasets and costs a second search for escalated queries. A development-selected 30% Zero contribution blended into Nano’s vector lowers Nano’s six-set score by 0.0051 (95% paired interval  $-0.0092$  to  $-0.0011$ ), consistent with how hard retrieval-strategy selection and query performance prediction remain [Arabzadeh et al., QPP].

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